Fault Protected High-Speed CAN FD Transceiver with Standby Mode

Features

- Meets the ISO 11898-2:2016 and SAE J2284-1 to SAE J2284-5 Physical Layer Standards
- Supports CAN FD and Data Rating up to 5 Mbps
- Short Propagation Delay Times and Fast Loop Times
- 5-V Power Supply, I/O Voltage Range Supports 2.8-V to 5.5-V MCU Interface
- Ultra-Low Current Standby Mode with Bus Wake-up Capability
- Ideal Passive Behavior to CAN Bus when Unpowered
- Common-Mode Input Voltage: ±30 V
- Protection Feature:
  - IEC 61000-4-2 ESD Protection up to ±15 kV
  - Bus Fault Protection: ±70 V
  - VCC and VIO (V variants only) Under-Voltage Protection
  - TXD Dominant Time-out Function and Bus-Dominant Time-out Function
  - Thermal Shutdown Protection
- Available in SOP8 Package and Leadless DFN3X3 Package

Applications

- All Devices Supporting Highly Loaded CAN Networks
- Field Industrial Automation, Sensor and Driver Systems
- Building, Security Control Systems
- Energy Storage Systems
- Telecom Base Station Status and Control

Description

The TPT1042 is a CAN transceiver that meets the ISO11898 high-speed CAN (Controller Area Network) physical layer standard. The device is designed to be used in CAN FD networks up to 5 Mbps, with enhanced timing margin and higher data rates in long and highly loaded networks. As designed, the device features crosswire, overvoltage, and loss of ground protection from −70 V to +70 V, over-temperature shutdown, with a −30 V to +30 V common-mode input voltage range. The TPT1042V has a secondary power supply input for I/O level shifting the input pin thresholds and RXD output level. This family has a low-current standby mode with CAN BUS waked-up capability. Additionally, all devices include many protection features to enhance the device and network robustness.

The TPT1042 and TPT1042V are available in SOP8 and DFN3X3-8L packages and are characterized from −40°C to +125°C.

Typical Application Circuit

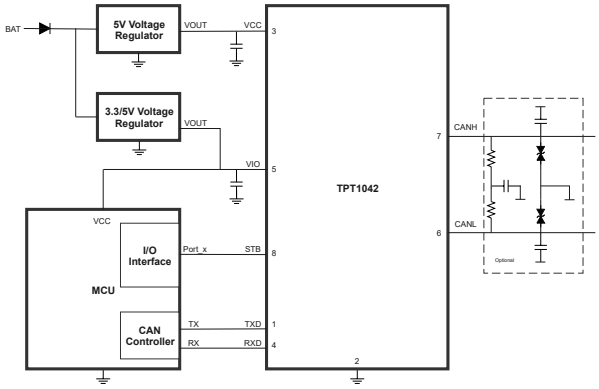


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Product Family Table

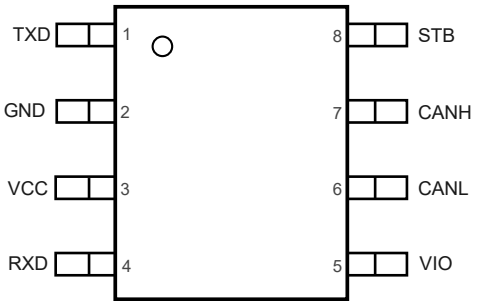
Order Number	VCC (V)	VIO (V)	BUS Protection (V)	Package
TPT1042V-SO1R-S	4.5 to 5.5	2.8 to 5.5	±70	SOP8
TPT1042V-DF6R-S	4.5 to 5.5	2.8 to 5.5	±70	DFN3X3-8L
TPT1042-SO1R-S	4.5 to 5.5	NC	±70	SOP8
TPT1042-DF6R-S	4.5 to 5.5	NC	±70	DFN3X3-8L

Revision History

Date	Revision	Notes
2020-12-05	Rev.Pre.0	Initial version
2022-11-20	Rev.A.0	Released version
2025-05-23	Rev.A.1	Corrected typos for the Conditons of the Standby Mode Supply Current parameter
2026-07-28	Rev.A.2	Updated the Functional Block Diagram The actual product remains unchanged.

Pin Configuration and Functions

TPT1042V
SOP8 Package
Top View



TPT1042V
DFN3X3-8 Package
Top View

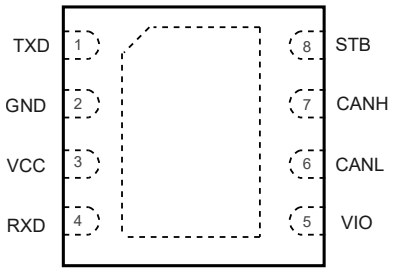
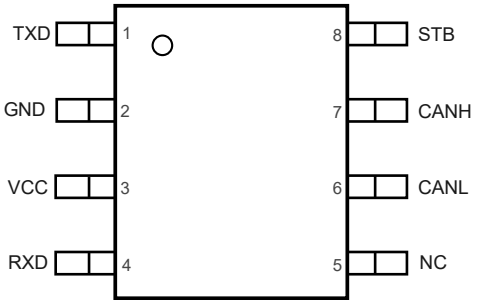


Table 1. Pin Functions: TPT1042VQ

Pin		I/O	Description
No.	Name		
1	TXD	I	CAN transmit data input (LOW for dominant and HIGH for recessive bus states)
2	GND	GND	Ground
3	VCC	POWER	Transceiver 5 V supply voltage
4	RXD	O	CAN receive data output (LOW for dominant and HIGH for recessive bus states)
5	VIO	POWER	Transceiver I/O level shifting supply voltage
6	CANL	BUS I/O	Low level CAN bus input/output line
7	CANH	BUS I/O	High level CAN bus Input/output line
8	STB	I	Standby Mode control input (active high)

TPT1042
SOP8 Package
Top View



TPT1042
DFN3X3-8 Package
Top View

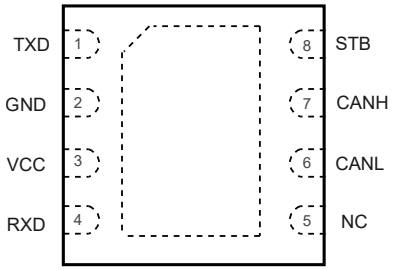


Table 2. Pin Functions: TPT1042Q

Pin		I/O	Description
No.	Name		
1	TXD	I	CAN transmit data input (LOW for dominant and HIGH for recessive bus states)
2	GND	GND	Ground
3	VCC	Power	Transceiver 5 V supply voltage
4	RXD	O	CAN receive data output (LOW for dominant and HIGH for recessive bus states)
5	NC	−	Not Connected
6	CANL	Bus I/O	Low-level CAN bus input/output line
7	CANH	Bus I/O	High-level CAN bus input/output line
8	STB	I	Standby Mode control input (active high)

Specifications

Absolute Maximum Ratings

Parameter		Min	Max	Unit
V_{CC}	5-V Bus Supply Voltage Range	−0.3	7	V
V_{IO}	I/O Level-Shifting Voltage Range	−0.3	7	V
V_{BUS}	CAN Bus I/O Voltage Range (CANH, CANL)	−70	70	V
V_{BUS_DIFF}	Differential Voltage of CAN Bus, CANH - CANL	−70	70	V
V_{LOGIC}	Logic Input and Output Terminal Voltage Range (TXD, STB, RXD)	−0.3	7	V
I_{O_RXD}	RXD (Receiver) Output Current	−8	8	mA
T_J	Maximum Junction Temperature	−40	150	°C
T_{STG}	Storage Temperature Range	−65	150	°C
T_{OTP}	Shutdown Junction Temperature		170	°C

(1) Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device. Exposure to any Absolute Maximum Rating condition for extended periods may affect device reliability and lifetime.
(2) This data was taken with the JEDEC low effective thermal conductivity test board.
(3) This data was taken with the JEDEC standard multilayer test boards.

ESD (Electrostatic Discharge Protection)

Parameter		Condition	Minimum Level	Unit
IEC	IEC Contact Discharge	IEC-61000-4-2, Bus Pin	±15	kV
	IEC Air-Gap Discharge	IEC-61000-4-2, Bus Pin	±15	kV
HBM	Human Body Model ESD	ANSI/ESDA/JEDEC JS-001, CAN Bus Pin	±8	kV
		ANSI/ESDA/JEDEC JS-001, All Pin	±7	kV
CDM	Charged Device Model ESD	ANSI/ESDA/JEDEC JS-002, All Pin	±1.5	kV
LU	Latch up	Latch up per JESD78, All Pin	±500	mA
Vtran	ISO7637-2 transients per IEC 62228-3, CANH, CANL	Pulse 1	-100	V
		Pulse 2a	75	V
		Pulse 3a	-150	V
		Pulse 3b	100	V

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

Recommended Operating Conditions

	Parameter	Min	Max	Unit
V_{IO}	Input/output voltage, TXD, RXD, STB	2.8	5.5	V
V_{CC}	Power Supply	4.5	5.5	V
$I_{OH(RXD)}$	RXD Terminal High-Level Output Current	−2		mA
$I_{OL(RXD)}$	RXD Terminal Low-Level Output Current		2	mA
T_A	Operating Ambient Temperature	−40	125	°C

Thermal Information

Package Type	θ_{JA}	θ_{JC}	Unit
SOP8	118	48	°C/W
DFN3x3-8	51	23	°C/W

Electrical Characteristics

All test conditions: V_{CC} = 4.5 V to 5.5 V, V_{IO} = 2.8 V to 5.5 V, R_L= 60 Ω, T = −40°C to 125°C, unless otherwise noted.

Parameter		Conditions	Min	Typ	Max	Unit
Pin V_{CC}, (Power supply)						
V_{CC}	Supply Voltage		4.5		5.5	V
V_{UV_VCC}	Rising Undervoltage Detection on V_{cc} for Protected Mode		-	4.1	4.5	V
	Falling Undervoltage Detection on V_{cc} for Protected Mode		3.6	3.9	-	V
V_{HYS_UVVCC}	Hysteresis Voltage on UV_{VCC} (1)		-	200	-	mV
I_{CC}	Standby Mode Supply Current	1042V, device with the "V" suffix, Standby Mode, V_{TXD} = V_{IO}, RL = 60 Ω, C_L = open, V_{STB} = V_{IO}	-	3.5	5	μA
		1042, device without the "V" suffix, Standby Mode, V_{TXD} = V_{CC}, RL = 60 Ω, C_L = open, V_{STB} = V_{CC}	-	10	20	μA
	Normal Mode Supply Current	Recessive, V_{TXD} = V_{CC}, R_L = 50 Ω, C_L = open, V_{STB} = 0 V	0.5	1.3	5	mA
		Dominant, V_{TXD} = 0 V, R_L = 60 Ω, C_L = open, V_{STB} = 0 V	20	50	80	mA
		Dominant bus fault, short circuit on bus lines, V_{TXD} = V_{STB} = 0 V, −3 V < (V_{CANH} = V_{CANL}) < +18V, R_L = C_L = open	2	60	110	mA
PIN V_{IO}, (I/O Level Adapter Supply) (2)						
V_{IO}	Supply Voltage on V_{io} Pin		2.8	-	5.5	V
V_{UV_VIO}	Rising Undervoltage Detection on V_{io} for Protected Mode		-	2	2.7	V
	Falling Undervoltage Detection on V_{io} for Protected Mode		1.3	1.9	-	V
V_{HYS_UVVIO}	Hysteresis Voltage on uv_{vio} for Protected Mode		-	100	-	mV
I_{IO}	Standby Mode Supply Current on Pin v_{io}	Standby Mode, RXD Floating, V_{TXD} = V_{STB} = V_{IO}	5	10	14	μA
	Normal Mode Supply Current on Pin v_{io}	Recessive, V_{STB} = 0 V, V_{TXD} = V_{IO}	-	15	30	μA

Parameter			Conditions	Min	Typ	Max	Unit
			Dominant, V_{STB} = 0 V, V_{TXD} = 0 V	-	200	600	μA
Pin STB, (Standby Mode Control Input)							
V_{IH}	High-Level Input Voltage			0.7 x V_{IO}	-	V_{IO}+ 0.3	V
V_{IL}	Low-Level Input Voltage			−0.3	-	0.3 x V_{IO}	V
I_{IH}	High-Level Input Current		V_{STB} = V_{CC} = V_{IO} = 5.5 V	−2	0	2	μA
I_{IL}	Low-Level Input Current		V_{STB} = 0 V, V_{CC} = V_{IO} = 5.5 V	−20	-11	-2	μA
$I_{lkg(OFF)}$	Unpowered Leakage Current		V_{STB} = 5.5 V, V_{CC} = V_{IO} = 0 V	−1	0	1	uA
Pin TXD, (CAN Transmit Data Input)							
V_{IH}	High-Level Input Voltage			0.7 x V_{IO}	-	V_{IO}+ 0.3	V
V_{IL}	Low-Level Input Voltage			−0.3	-	0.3 x V_{IO}	V
I_{IH}	High-Level Input Current		V_{TXD} = V_{CC} = V_{IO} =5.5 V	−2	0	2	μA
I_{IL}	Low-Level Input Current		V_{TXD} =0 V, V_{CC} = V_{IO} = 5.5 V	−250	−160	−30	μA
I_{LKG_OFF}	Unpowered Leakage Current		V_{TXD} = 5.5 V, V_{CC} = V_{IO}= 5.5 V	−1	0	1	μA
C_I	Input Capacitance (1)			-	5	10	pF
Pin RXD, (CAN Receive Data Output)							
I_{OH}	High-Level Output Current		VRXD = VIO − 0.4 V	−8	−6	−1	mA
I_{OL}	Low-Level Output Current		VRXD = 0.4 V, bus dominant	2	5	12	mA
I_{LKG_OFF}	Unpowered Leakage Current		V_{RXD} = 5.5 V, V_{CC} = V_{IO} = 0 V	−1	0	1	μA
Pins CANH and CANL, (CAN Bus Lines)							
V_{O_DOM}	Dominant Bus Output Voltage	CANH	V_{TXD} = 0 V, V_{STB} = 0 V, 50 Ω ≤ R_L ≤ 65 Ω, C_L = open, t < $t_{to(dom)TXD}$	2.75	3.5	4.5	V
		CANL		0.5	1.5	2.25	V
V_{SYM_DC}	DC Output Symmetry (dominant or recessive) (V_{CC} – $V_{O(CANH)}$ – $V_{O(CANL)}$)		V_{STB} = 0 V, R_L = 60 Ω, C_L = open,	−0.4	-	0.84	V
V_{SYM}	Transient Symmetry (dominant or recessive) ($V_{O(CANH)}$ + $V_{O(CANL)}$) / V_{CC} (1)		4.75V ≤ V_{CC} ≤ 5.25 V, V_{STB} = 0 V, R_L = 60 Ω, C_{SPLIT} = 4.7 nF, C_L = open, T_{XD} = 250 kHz, 1 MHz	0.9	1	1.1	V/V
V_{OD_DOM}	Differential Output Voltage (dominant)		Normal mode, t < $t_{to(dom)TXD}$, V_{TXD} = 0 V, V_{STB} = 0 V,	1.5	-	3	V

Parameter			Conditions	Min	Typ	Max	Unit
			4.75V ≤ V_{CC} ≤ 5.25 V, 45 Ω ≤ R_L < 50 Ω, C_L = open				
			Normal mode, t < $t_{to(dom)TXD}$,V_{TXD} = 0 V, V_{STB} = 0 V, 4.75V ≤ V_{CC} ≤ 5.25 V, 50 Ω ≤ R_L < 65 Ω, C_L = open	1.5	-	3	V
			Normal mode, t < $t_{to(dom)TXD}$, V_{TXD} = 0 V, V_{STB} = 0 V, 4.75V ≤ V_{CC} ≤ 5.25 V, 65 Ω ≤ R_L ≤ 70 Ω, C_L = open	1.5	-	3.3	V
			Normal mode, t < $t_{to(dom)TXD}$, V_{TXD} = 0 V, V_{STB} = 0 V, 4.75V ≤ V_{CC} ≤ 5.25 V, R_L = 2240 Ω, C_L = open	1.5	-	5	V
V_{OD_REC}	Differential Output Voltage (recessive)		Normal mode, V_{TXD} = V_{CC}, V_{STB} = 0 V, no load	−50	-	50	mV
			Standby mode, V_{TXD} = V_{STB} = V_{CC}, no load	−150	-	150	mV
V_{O_REC}	Recessive Bus Output Voltage		Normal mode, V_{TXD} = V_{IO} = V_{CC}, V_{STB} = 0 V, no load	2	0.5 x VCC	3	V
			Standby mode, V_{TXD} = V_{STB} = V_{IO} = V_{CC}, no load	−0.1	-	0.1	V
V_{CM}	Common Mode Range		V_{STB} = 0 or V_{CC}or V_{IO}	−30	-	30	V
$V_{TH_RX_DIF}$	Differential Receiver Threshold Voltage		Normal mode, V_{STB} = 0 V, −30 V ≤ V_{CANH}/ V_{CANL} ≤ 30 V	0.5	0.7	0.9	V
			Standby mode, V_{STB} = V_{IO}, −30 V ≤ V_{CANH}/ V_{CANL} ≤ 30 V	0.4	0.7	1.15	V
V_{REC_RX}	Receiver Recessive Voltage, Normal Mode		−30 V ≤ V_{CANH}/V_{CANL} ≤ 30 V	−4	-	0.5	V
	Receiver Recessive Voltage, Standby Mode			−4	-	0.4	V
V_{DOM_RX}	Receiver Dominant Voltage, Normal Mode		−30 V ≤V_{CANH}/V_{CANL} ≤ 30 V	0.9	-	9	V
	Receiver Dominant Voltage, Standby Mode			1.15	-	9	V
$V_{HYS_RX_DIF}$	Differential Receiver Hysteresis Voltage		Normal mode, −30 V ≤ V_{CANH}/ V_{CANL} ≤ 30 V	50	120	200	mV
			Standby mode, −30 V ≤ V_{CANH}/ V_{CANL} ≤ 30 V	20	70	120	mV
$I_{O_SC_DOM}$	Dominant Short-Circuit Output Current	CANH	V_{STB} = 0 V, V_{CANH} = −15 V to 18 V, CANL = open, V_{TXD} = 0 V	−115	-	-	mA

Parameter			Conditions	Min	Typ	Max	Unit
		CANL	V_{STB} = 0 V, V_{CANL} = −15 V to 18 V, CANH = open, V_{TXD} = 0 V	-	-	115	mA
$I_{O_SC_REC}$	Recessive Short-Circuit Output Current		−27 V ≤ V_{CANH}/ V_{CANL} ≤ 32 V, V_{TXD} = V_{CC}, normal modes	−5	-	5	mA
I_{LKG_IOFF}	Power-off (unpowered) Bus Input Leakage Current		V_{CC} = V_{IO} = 0 V or V_{CC} = V_{IO} shorted to ground via 47kΩ, V_{CANH} = 5 V, V_{CANL} = 5 V	−5	-	5	μA
R_{IN}	Input Resistance (CANH or CANL)		V_{TXD} = V_{CC} = V_{IO} = 5 V, V_{STB} = 0 V, −30 V ≤ V_{CM} ≤ +30 V	10	23	35	kΩ
R_{IN_M}	Input Resistance Matching: [1 − $R_{IN(CANH)}$ / $R_{IN(CANL)}$] × 100%		V_{TXD} = V_{CC} = V_{IO} = 5 V, V_{STB} =0 V	−2	-	2	%
			V_{CANH} = V_{CANL} =5 V				
R_{ID}	Differential Input Resistance		V_{TXD} = V_{CC} = V_{IO} = 5 V, V_{STB} = 0 V, −30 V ≤ V_{CM} ≤ +30 V	30	47	60	kΩ
C_I	Input Capacitance to Ground (CANH or CANL) (1)			-	-	20	pF
C_{ID}	Differential Input Capacitance (1)			-	-	10	pF

(1) The data is based on bench test and design simulation.
(2) Only device with V suffix (TPT1042V) have a V_{IO} Pin, device without V suffix (TPT1042) V_{IO} connoted to V_{CC} internally.

AC Timing Requirements

All test conditions: V_{CC} = 4.5 V to 5.5 V, V_{IO} = 2.8 V to 5.5 V, R_L= 60 Ω, T_A = −40°C to 125°C, unless otherwise noted.

Parameter		Conditions	Min	Typ	Max	Unit
Transceiver Switching Characteristics						
t_{pLD}	Propagation delay time, low TXD to driver dominant (recessive to dominant)	Normal mode, V_{STB} = 0 V, R_L = 60 Ω, C_L = 100 pF	-	60	100	ns
t_{pHR}	Propagation delay time, high TXD to driver recessive (dominant to recessive)		-	60	100	ns
t_{SK_P}	Pulse Skew (|t_{pHR}− t_{pLD}|)		-	10	35	ns
t_R	Differential Output Signal Rise Time (1)		-	45	-	ns
t_F	Differential Output Signal Fall Time (1)		-	45	-	ns
$t_{PROP_TXDL-RXDL}$	Total loop delay, driver input (TXD) low to receiver output	Normal mode, V_{STB} = 0 V,	-	110	220	ns

Parameter		Conditions	Min	Typ	Max	Unit
	(RXD) low, recessive to dominant	R_L = 60 Ω, C_L = 100 pF, $C_{L(RXD)}$ = 15 pF,				
$t_{PROP_TXDH-RXDH}$	Total loop delay, driver input (TXD) high to receiver output (RXD) high, dominant to recessive		-	140	220	ns
t_{pRH}	Propagation Delay Time, Bus Recessive Input to RXD High Output (Dominant to Recessive) (1)	V_{STB} = 0 V, $C_{L(RXD)}$ = 15 pF	-	90	120	ns
t_{pDL}	Propagation Delay Time, Bus Dominant Input to RXD Low Output (Recessive to Dominant) (1)		-	90	120	ns
t_{R_R}	RXD Output Signal Rise Time (1)		-	20	-	ns
t_{R_F}	RXD Output Signal Fall Time (1)		-	20	-	ns
FD Timing Parameters						
t_{BIT_BUS}	Bit time on CAN bus output pins with $t_{BIT(TXD)}$ = 500 ns(1)	V_{STB} = 0 V, R_L = 60 Ω, C_L = 100 pF, $C_{L(RXD)}$ = 15 pF, Δt_{REC} = $t_{BIT(RXD)}$ − $t_{BIT(BUS)}$	435	-	530	ns
	Bit time on CAN bus output pins with $t_{BIT(TXD)}$ = 200 ns(1)		155	-	210	ns
t_{BIT_RXD}	Bit time on RXD output pins with $t_{BIT(TXD)}$ = 500 ns		400	-	550	ns
	Bit time on RXD output pins with $t_{BIT(TXD)}$ = 200 ns		120	-	220	ns
Δt_{REC}	Receiver timing symmetry with $t_{BIT(TXD)}$ = 500 ns(1)		−65	-	40	ns
	Receiver timing symmetry with $t_{BIT(TXD)}$ = 200 ns(1)		−45	-	15	ns
Device Timing Parameters						
t_{TXD_DTO}	TXD dominant time-out time	normal mode, V_{STB} = 0 V, R_L = 60 Ω, C_L = open, V_{TXD} = 0 V	0.3	2	5	ms
t_{BUS_DTO}	Bus dominant time-out time	Standby mode	0.3	2	5	ms
$t_{BUS_WAKE_FILTER}$	Bus wake-up filter time	Standby mode	0.5	1.5	5	μs
t_{dMODE}	Standby to normal mode delay time		3	8	50	μs

(1) The data is based on bench test and design simulation.

Parameter Measurement Information

Test Circuit

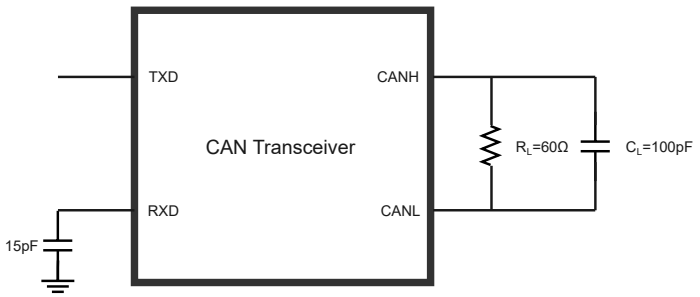


Figure 1. CAN transceiver timing parameter test circuit

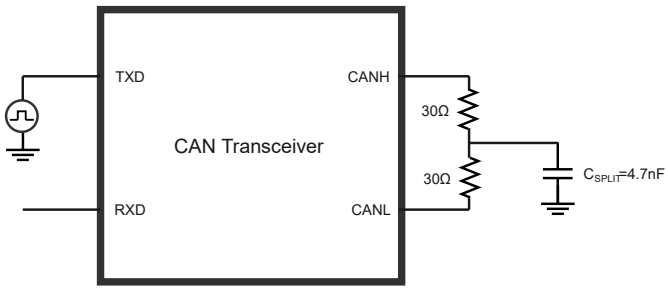


Figure 2. CAN transceiver driver symmetry test circuit

Parameter Diagram

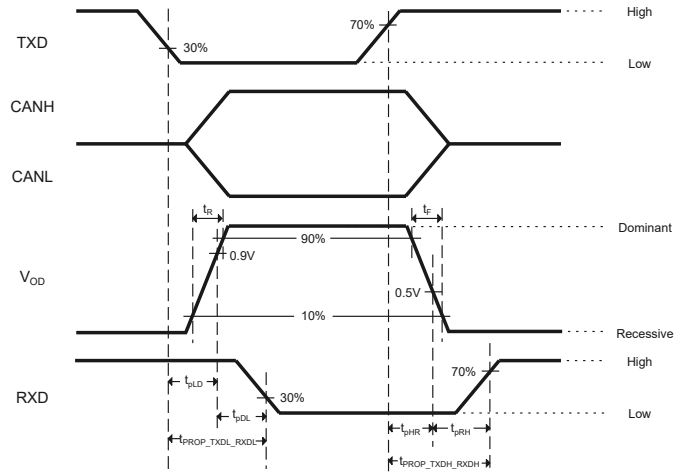


Figure 3. CAN transceiver timing diagram

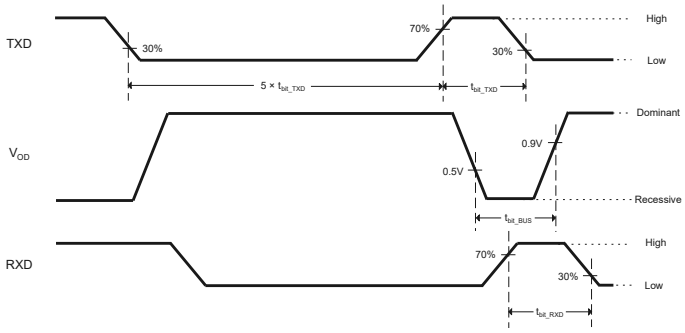


Figure 4. CAN FD timing parameter diagram

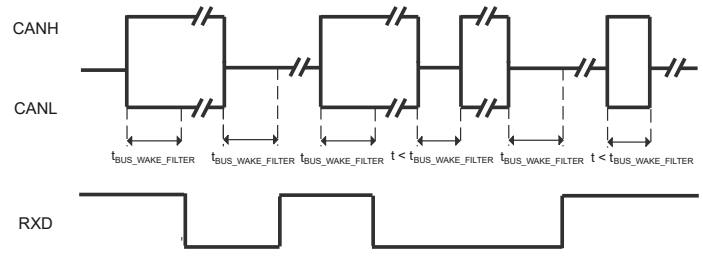


Figure 5. Wake-up timing diagram

Detailed Description

Overview

The TPT1042 is a CAN transceiver that meets the ISO11898 high-speed CAN (Controller Area Network) physical layer standard. The device is designed to be used in CAN FD networks up to 5 Mbps, with enhanced timing margin and higher data rates in long and highly loaded networks. As designed, the device features crosswire, overvoltage, and loss of ground protection from −70 V to +70 V, over-temperature shutdown, with a −30 V to +30 V common-mode input voltage range. The TPT1042V has a secondary power supply input for I/O level shifting the input pin thresholds and RXD output level. This family has a low-current standby mode with CAN BUS waked-up capability. Additionally, all devices include many protection features to enhance the device and network robustness.

Functional Block Diagram

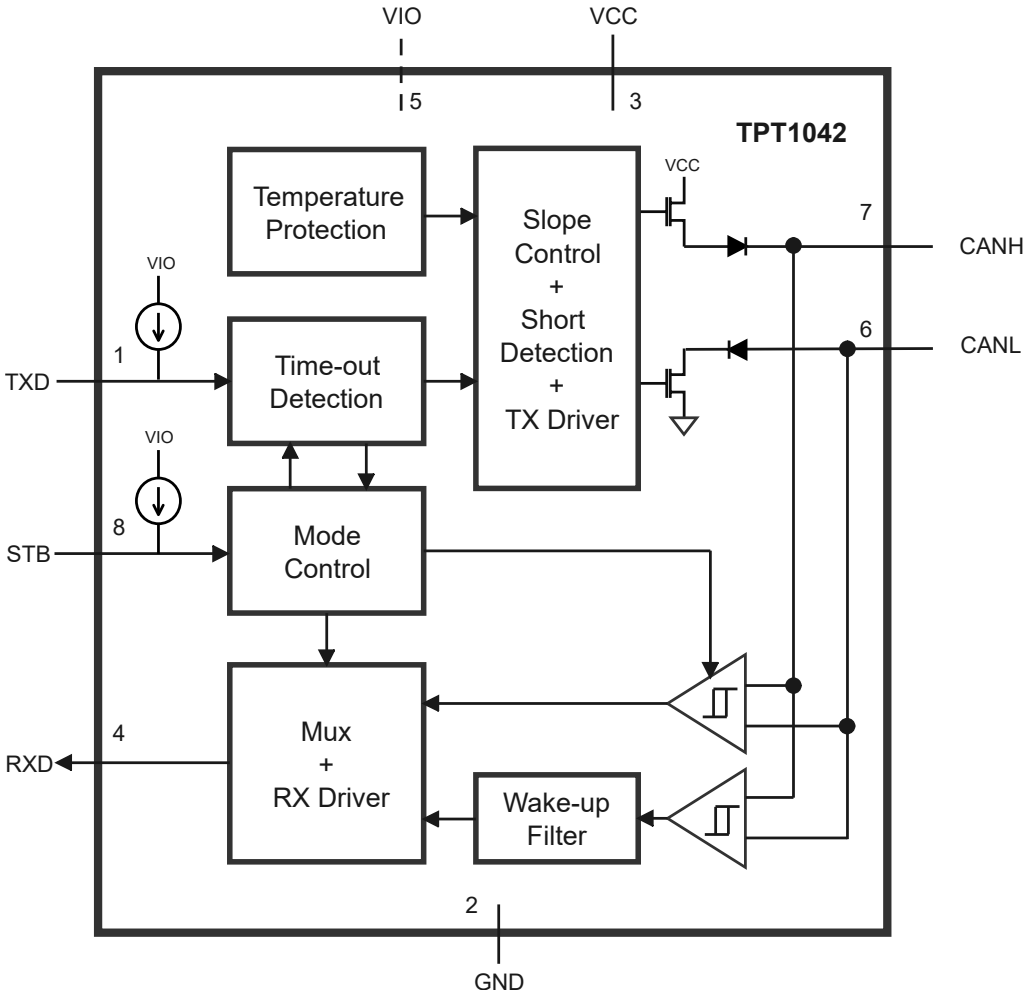


Figure 6. Functional Block Diagram

Feature Description

Table 3. Driver Function Table

Device	Inputs		Outputs		Driven BUS State
	STB	TXD	CANH	CANL	
All Devices	L	L	H	L	Dominant
		H or Open	Z	Z	Recessive
	H or Open	X	Z	Z	Recessive

Table 4. Receiver Function Table

Device Mode	CAN Differential Inputs V_{ID} = V_{CANH} − V_{CANL}	Bus State	RXD Terminal
Normal or Standby	$V_{ID} \geq V_{IT+(MAX)}$	Dominant	L
	$V_{IT-(MIN)} < V_{ID} < V_{IT+(MAX)}$	Indeterminate	Indeterminate
	$V_{ID} \leq V_{IT-(MIN)}$	Recessive	H
	Open ($V_{ID} \approx 0$ V)	Open	H

Device Operating Modes

Normal Mode

A low level on the STB pin selects the normal mode. In this mode, the transceiver will transmit and receive data via the bus lines CANH and CANL. The differential receiver converts the analog data on the bus lines into digital data, which is output to the RXD pin. The slopes of the output signals on the bus lines are controlled internally and optimized to guarantee the lowest possibility for Electro Magnetic Emission (EME).

Standby Mode

Activate the low-power standby mode by setting the STB terminal high. In this mode, the bus transmitter will not send data, nor will the normal mode receiver accept data as the bus lines are biased to ground minimizing the system supply current. Only the low-power receiver will be actively monitoring the bus for activity. RXD indicates a valid wake-up event after a valid wake-up signal has been detected on the Bus. The low-power receiver is powered using the VIO pin only. This allows V_{CC} to be removed reducing power consumption further.

Device Local Faults

TXD Dominant Time-out

The device will detect TXD dominant time-out and prevents a permanent low on the TXD pin drive CAN bus into permanent dominant blocking the CAN bus network. If the TXD remains low for longer than $t_{TXD_DTO,}$ the transmitter will be disabled until the fault flag has been cleared. The TXD dominant time-out time also defines that the data rate should be faster than 40 kbit/s.

Bus Dominant Time-out

The device will detect Bus dominant time-out and prevent a permanent dominant on the CAN bus blocking the CAN bus network. The fault flag is set if the CAN bus remains dominant for longer than t_{BUS_DTO}, and is released as soon as the bus returns to recessive.

Under-Voltage Lockout (UVLO)

The device integrated under-voltage detect and lockout circuit of the supply terminal to keep the device in the protected mode if the supply voltage drops below the threshold until the supply voltage is higher than the UVLO threshold. This protects the device and system during the under-voltage event on supply terminals.

Over-Temperature Protection (OTP)

The output drivers are protected against over-temperature conditions. If the virtual junction temperature exceeds the shutdown junction temperature T_{OTP}, the output drivers will be disabled until the virtual junction temperature falls below T_{OTP} and TXD becomes recessive again. Including the TXD condition minimizes output driver oscillation due to temperature drift.

Application and Implementation

> Note
>
> Information in the following application sections is not part of the 3PEAK's component specification and 3PEAK does not warrant its accuracy or completeness. 3PEAK's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

Application Information

The TPT1042 is a CAN transceiver to support CAN FD function up to 5 Mbps, with BUS protection voltage from –70 V to +70 V, overtemperature shutdown, a −30 V to +30 V common-mode range. The VIO of TPT1042 can support the voltage level of TXD and RXD from 2.8 V to 5.5 V. The following sections show a typical application of the TPT1042.

Typical Application

Figure 7 shows the typical application schematic of the TPT1042.

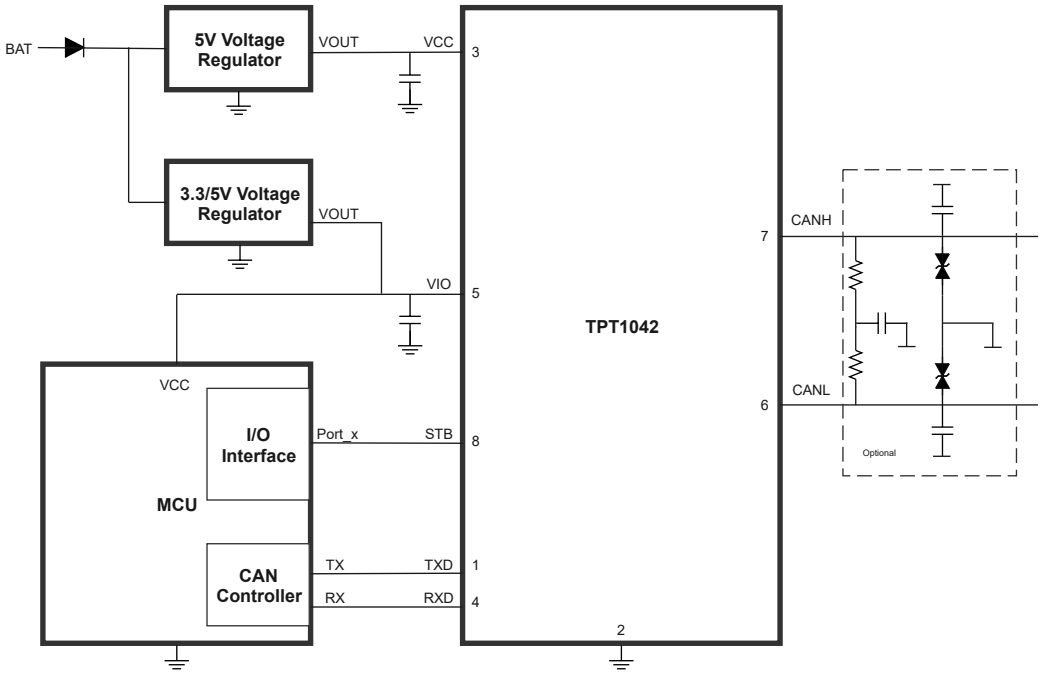
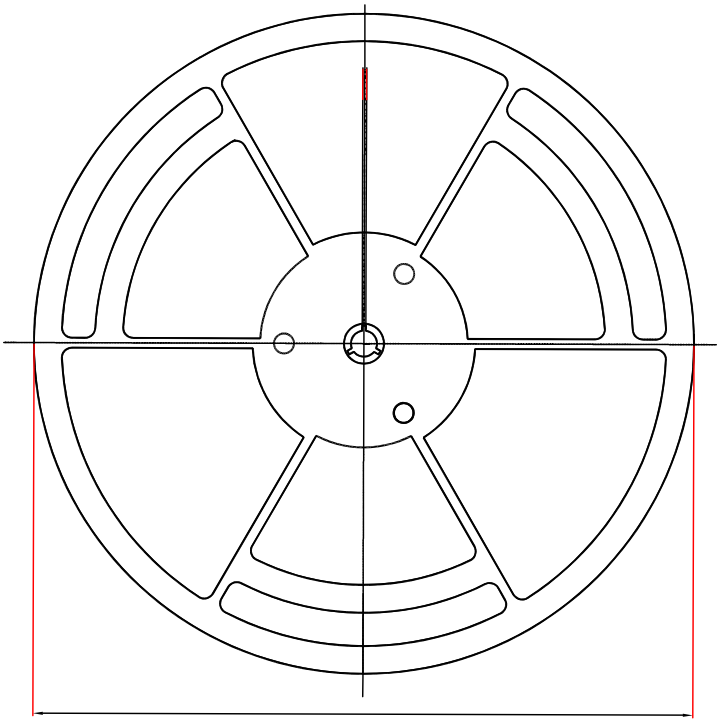
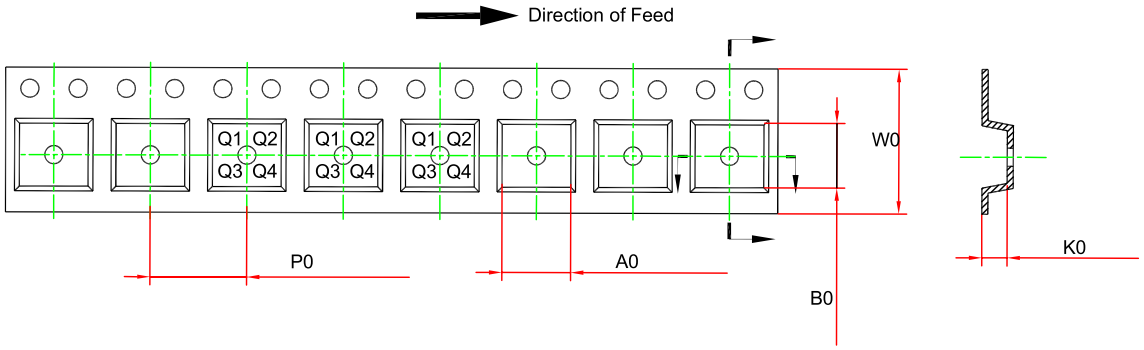


Figure 7. Typical Application Circuit

Tape and Reel Information



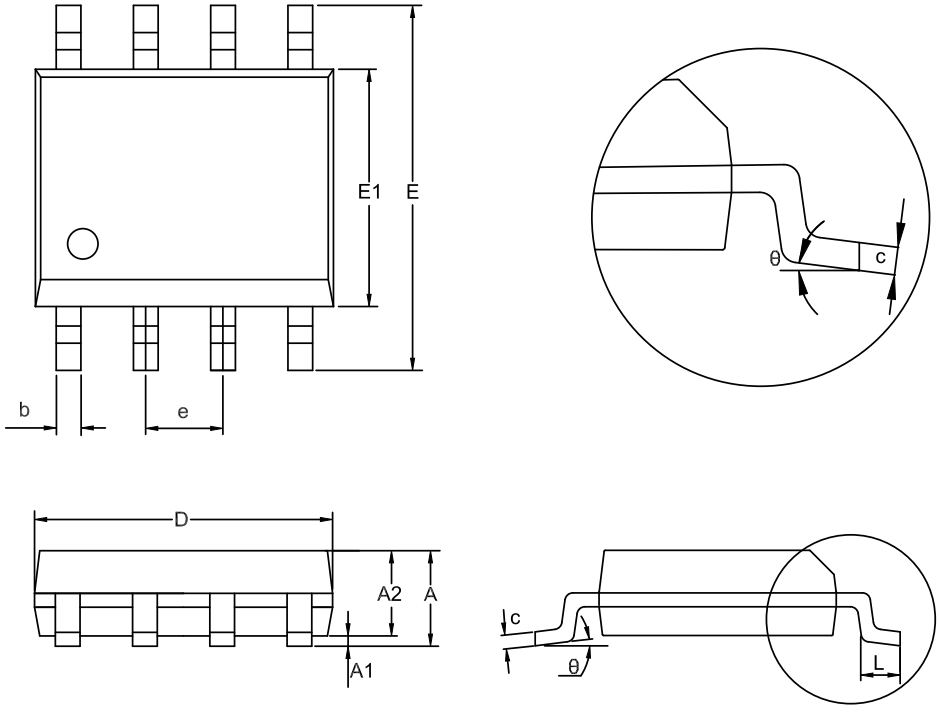


Order Number	Package	D1 (mm)	W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P0 (mm)	W0 (mm)	Pin1 Quadrant
TPT1042V-SO1R-S	SOP8	330	17.6	6.5	5.4	2.0	8.0	12.0	Q1
TPT1042V-DF6R-S	DFN3x3-8	330	17.6	3.3	3.3	1.1	8.0	12.0	Q1
TPT1042-SO1R-S	SOP8	330	17.6	6.5	5.4	2.0	8.0	12.0	Q1
TPT1042-DF6R-S	DFN3x3-8	330	17.6	3.3	3.3	1.1	8.0	12.0	Q1

Package Outline Dimensions

SOP8

Package Outline Dimensions SO1(SOP-8-A)



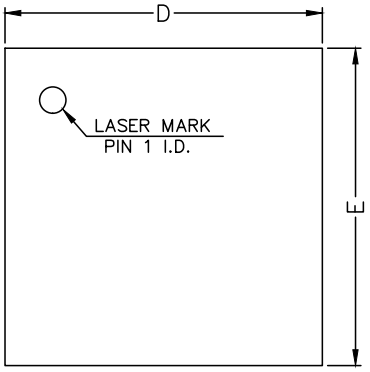
Symbol	Dimensions In Millimeters		Dimensions In Inches	
	MIN	MAX	MIN	MAX
A	1.350	1.750	0.053	0.069
A1	0.050	0.250	0.002	0.010
A2	1.250	1.550	0.049	0.061
b	0.330	0.510	0.013	0.020
c	0.170	0.250	0.007	0.010
D	4.700	5.100	0.185	0.201
E	5.800	6.200	0.228	0.244
E1	3.800	4.000	0.150	0.157
e	1.270 BSC		0.050 BSC	
L	0.400	1.000	0.016	0.039
θ	0	8°	0	8°

NOTES

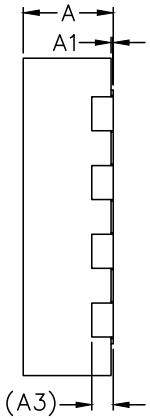
1. Do not include mold flash or protrusion.
2. This drawing is subject to change without notice.

DFN3X3-8

Package Outline Dimensions DF6 (DFN3X3-8-E)



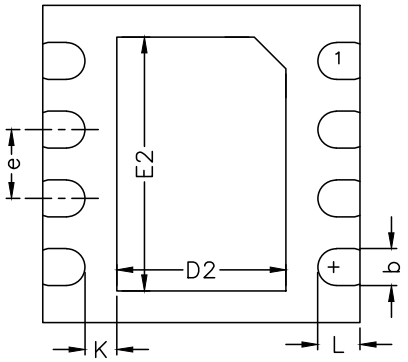
TOP VIEW



SIDE VIEW



SIDE VIEW



BOTTOM VIEW

NOTES

1. Do not include mold flash or protrusion.
2. This drawing is subject to change without notice.

Symbol	Dimensions In Millimeters		Dimensions In Inches	
	MIN	MAX	MIN	MAX
A	0.800	0.900	0.031	0.035
A1	0.000	0.050	0.000	0.002
b	0.250	0.400	0.010	0.016
A3	0.150	0.250	0.006	0.010
D	2.900	3.100	0.114	0.122
D2	1.500	1.700	0.059	0.067
E	2.900	3.100	0.114	0.122
E2	2.300	2.500	0.091	0.098
e	0.650 BSC		0.026 BSC	
L	0.300	0.500	0.012	0.020

Order Information

Order Number	Operating Temperature Range	Package	Marking Information	MSL	Transport Media, Quantity	Eco Plan
TPT1042V-SO1R-S	−40 to 125°C	SOP8	1042V	MSL3	Tape and Reel, 4000	Green
TPT1042V-DF6R-S	−40 to 125°C	DFN3x3-8	1042V	MSL3	Tape and Reel, 4000	Green
TPT1042-SO1R-S	−40 to 125°C	SOP8	T1042	MSL3	Tape and Reel, 4000	Green
TPT1042-DF6R-S	−40 to 125°C	DFN3x3-8	T1042	MSL3	Tape and Reel, 4000	Green

Green: 3PEAK defines "Green" to mean RoHS compatible and free of halogen substances.

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